

Karthick N

Senior QA Engineer / SDET — Python, Playwright, API, CI/CD

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Professional Summary

Senior QA Engineer / SDET with 4.5+ years designing and scaling Python-based test automation frameworks for SaaS and enterprise applications. Experienced in end-to-end UI, API, functional, and regression testing across web, desktop, and backend systems. Skilled in Playwright, Pytest, REST API automation, CI/CD integration, performance testing, and data validation within Agile/Scrum teams.

Technical Skills

Languages & Test Automation: Python, SQL, Playwright, Pytest, Pywinauto, Test Automation Frameworks

Testing & API: Functional, Regression, End-to-End, Smoke, Sanity, UI/API Testing, REST API, Postman, API Automation

Framework & Strategy: Test Automation Strategy, Test Framework Design, Data-Driven Testing

Performance & CI/CD: Locust, Load Testing, Performance Validation, Jenkins, Continuous Integration, CI/CD, Git

Cloud & Data: AWS, Pandas, Excel Automation, CSV/Excel Processing, Data Validation, Data Migration, VBA

Agile & Tools: JIRA, QA Touch, Agile, Scrum, Defect Tracking, Reporting, Process Optimization

Professional Experience

Senior QA Engineer (Data & Automation)

Mako IT Lab · Oct 2025 – Present

- Engineered Python + Pandas validation pipelines for large-scale system migrations, processing 100K+ records across CSV/Excel datasets with automated reconciliation against source-of-truth REST APIs.
- Automated bulk REST API operations (create, update, sync) cutting manual data-entry effort by 80% and eliminating cross-system data drift during enterprise go-lives.
- Built workflow validation suites that surface data-integrity defects pre-cutover, reducing post-migration hotfix tickets and improving release confidence.

Software Quality Assurance Engineer

Freightify — SaaS platform for global freight pricing · Apr 2023 – Sep 2025

- Architected a Python + Pytest test automation framework with modular fixtures and data-driven testing; scaled to 1000+ automated regression test cases covering pricing, rate-management, and quote workflows across a multi-tenant SaaS platform.
- Owned end-to-end functional, regression, and API test coverage across web, desktop, and backend services; investigated test failures via log analysis and partnered with developers to drive defects to resolution.
- Integrated the suite into Jenkins CI/CD and Continuous Integration pipelines as a release gate, reducing regression cycle time from 2 days to under 4 hours and catching breakages before merge.
- Led a performance-testing POC with Locust and Postman, profiling backend response times under concurrent load and surfacing 3 latency hotspots routed back to engineering.
- Collaborated with PMs and developers in Agile/Scrum sprints to define acceptance criteria, review user stories, track defects, and estimate test effort during sprint planning and grooming.

Software Test Engineer

Freightify · Jan 2022 – Mar 2023

- Designed and executed UI, API, functional, regression, and end-to-end test cases across web, mobile, and desktop modules, helping establish the foundation of the team's automation strategy.
- Built API-level Python automation that replaced manual validation for high-volume rate-update workflows, lifting automated regression coverage and shortening release verification.
- Automated Excel-based file-processing flows using Pandas and developed Pywinauto scripts for legacy desktop application testing.

Data Analyst

Freightify — Includes 6-month internship (Mar – Aug 2021) · Mar 2021 – Dec 2021

- Standardized and audited carrier rate sheets in Excel, resolving recurring data-quality issues and improving downstream system accuracy.
- Optimized daily processing workflows and delivered executive-facing reports while meeting strict SLAs.

Professional Skills

Test Strategy & Framework Design: Design scalable, maintainable test automation strategies and modular test frameworks for enterprise applications.

Defect Analysis & Root Cause Investigation: Systematic debugging across UI, API, and data layers using log analysis, troubleshooting, and defect tracking.

Stakeholder Collaboration: Translate ambiguous requirements into testable acceptance criteria with PMs, developers, and business teams; comfortable working across time zones.

Ownership & Continuous Learning: Drive quality improvements end-to-end and rapidly adopt new frameworks, domains, and testing tools.

Education

University of Madras · B.Com in Corporate Secretarialship — First Class · 2017 – 2020